

MULLER: A Multimodal Data Lake Format for Collaborative AI Data Workflows

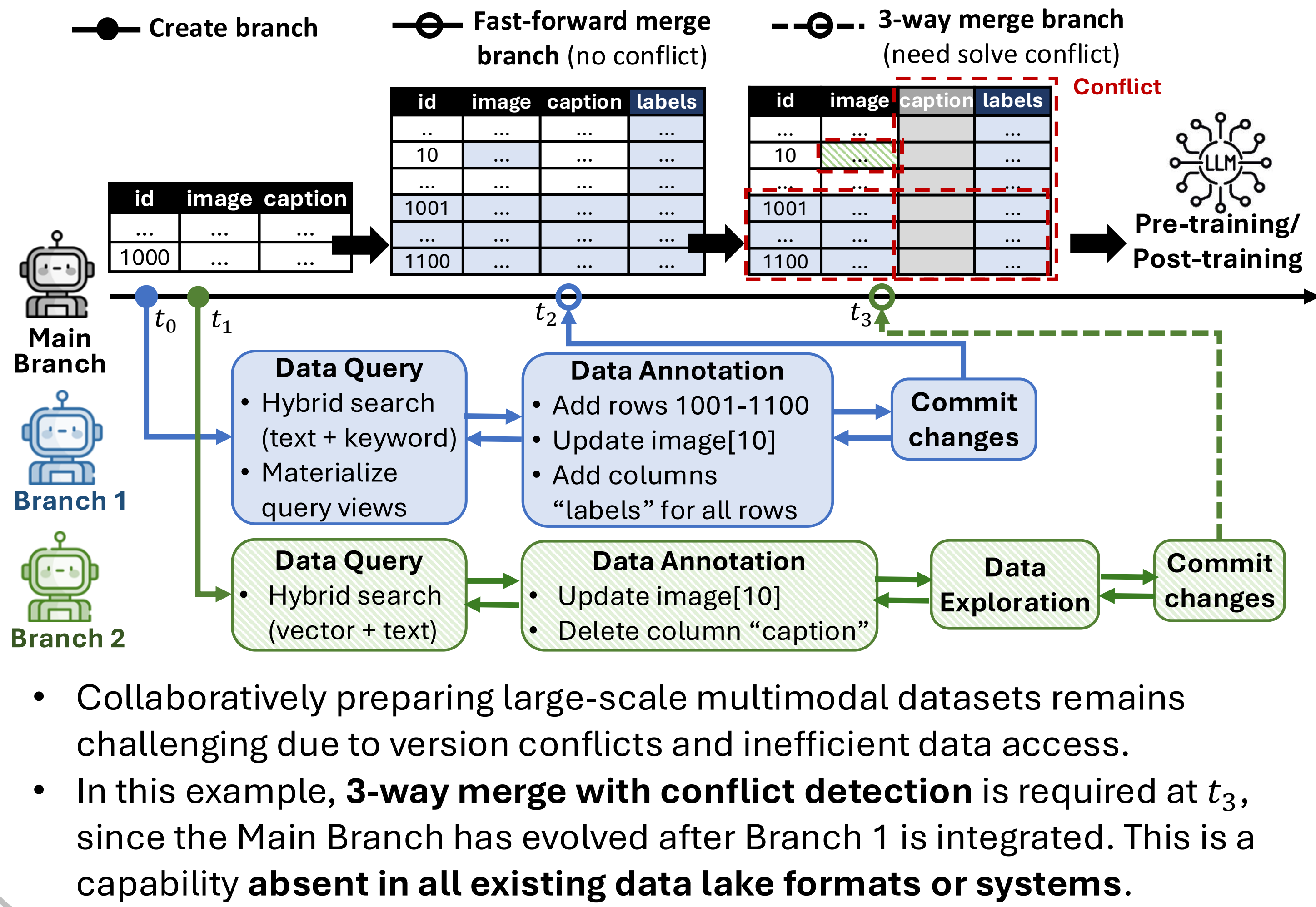
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Motivation

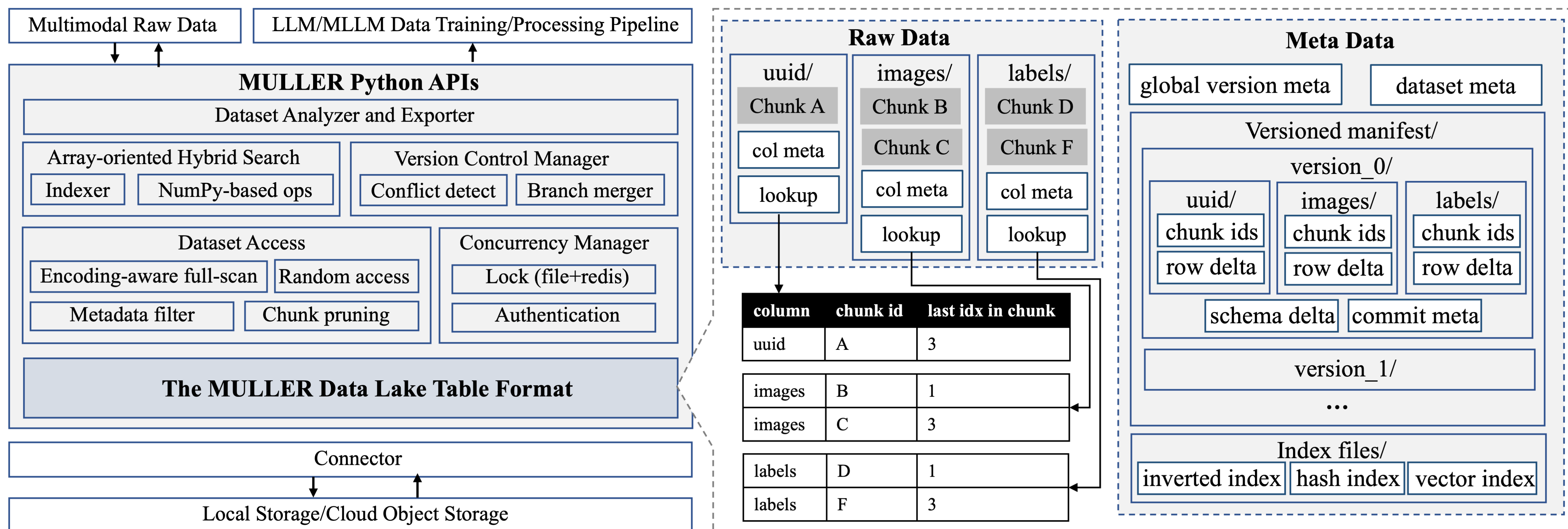


Comparison with State-of-the-Arts

	Lance*	Parquet*	F3*	Deep Lake†	Iceberg†	MULLER*
Multimodal data support	✓	X	X	✓	X	✓
Efficient random access	✓	X	✓	X	X	✓
Efficient full scan	✓	✓	N.A.	✓	✓	✓
Hybrid data query on vector, text, and scalar	✓	X	X	✓	X	✓
Branch-isolated schema evolution	X	X	X	✓	X	✓
Advanced Git-like versioning (e.g., three-way merge)	X	X	X	X	X	✓
Integration with LLM/MLLM data workflows	✓	X	X	✓	X	✓
Python API	✓	✓	X	✓	Pylceberg	✓

- Evaluated based on official documentation and experiments.
- Existing data management solutions, including different **data lake formats*** and **systems†**, lack a unified abstraction for
 - AI Data workflows over heterogeneous modalities;
 - fast random access for index-to-record lookups;
 - branch-isolated schema evolution and row-level conflict detection.

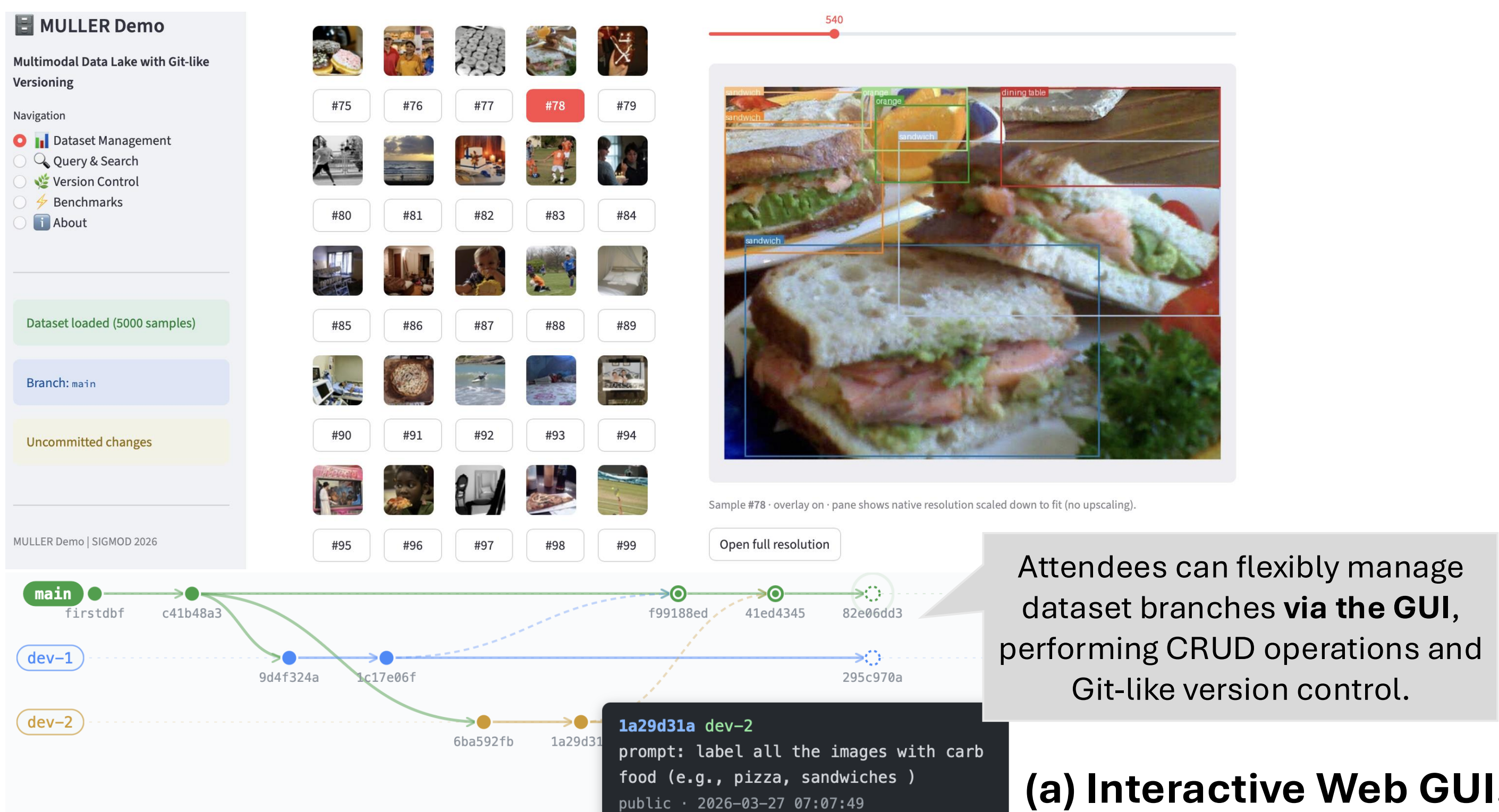
System Architecture & Data Format



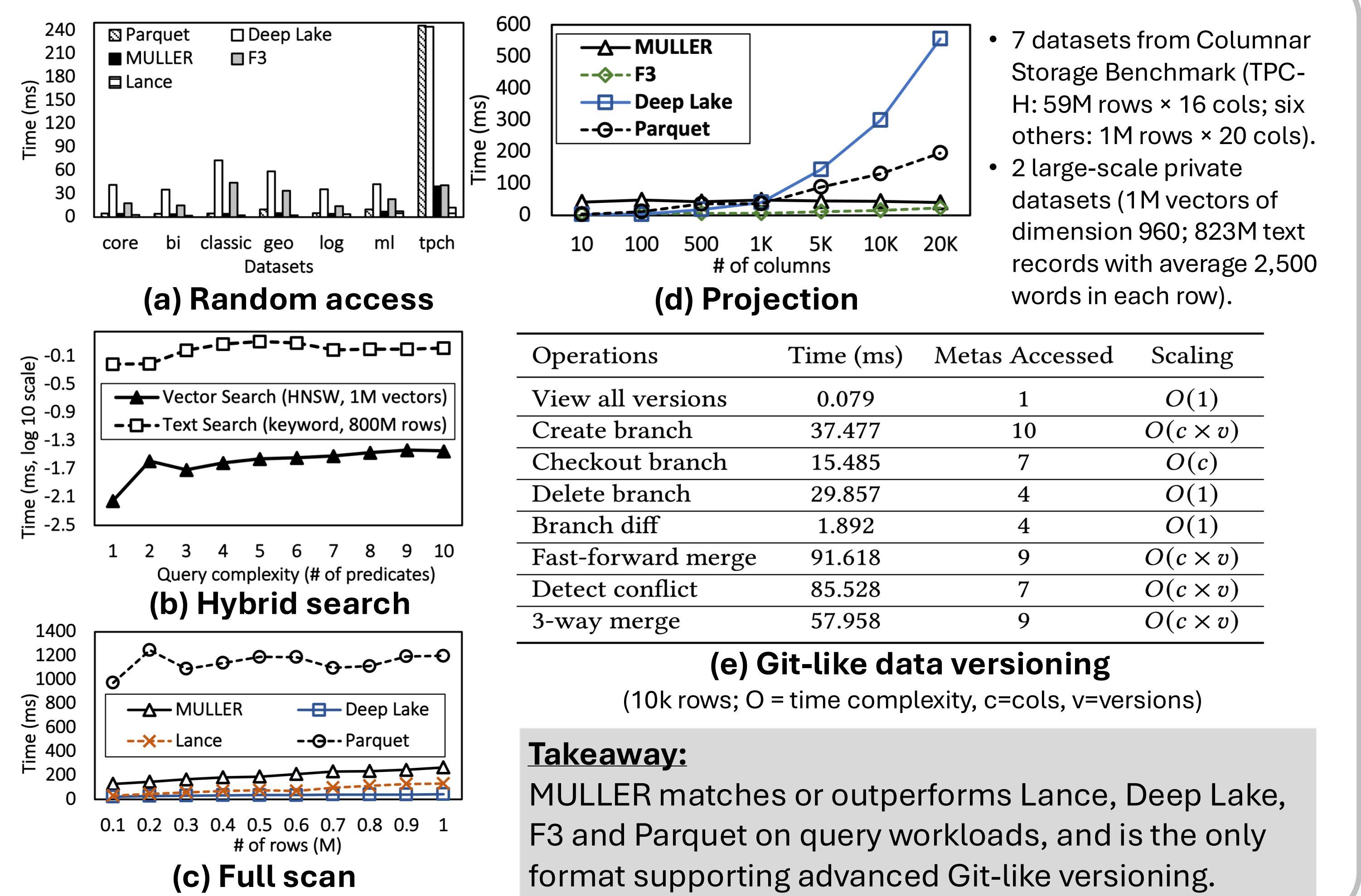
- MULLER adopts a disaggregated compute-storage design, while exposing ML-friendly Python APIs. The innovative design includes:
- Data Lake Format:**
- Versioned manifest for Git-like data management
 - Logarithmic data localization with constant-time access
- Core Functions:**
- Array-oriented hybrid search engine
 - Version control manager
 - Concurrency manager

Demonstration Scenarios

We offer three interfaces to guide the attendees through the **collaborative multimodal AI dataset lifecycle**, i.e., multimodal dataset construction, collaborative data annotation, data exploration, data query and data export.



Performance Evaluation



More Details



GitHub



Documents & API References



Video demo (Python APIs)



Video demo (Web GUI)